

Tutorial #7

Topics covered:

- Connectedness

Problem 1

Let $\mathcal{T}$ and $\mathcal{T}'$ be two topologies on X . If $\mathcal{T}' \supset \mathcal{T}$, what does connectedness of X in one topology imply about connectedness in the other?

Problem 2

Let $\{A_n\}$ be a sequence of connected subspaces of X , such that $A_n \cap A_{n+1} \neq \emptyset$ for all n . Show that $\bigcup A_n$ is connected.

Problem 3

Let $\{A_\alpha\}$ be a collection of connected subspaces of X ; let A be a connected subspace of X . Show that if $A \cap A_\alpha \neq \emptyset$ for all α , then $A \cup (\bigcup A_\alpha)$ is connected.

Problem 4

Show that if X is an infinite set, it is connected in the finite complement topology.

Problem 5

Let $A \subset X$. Show that if C is a connected subspace of X that intersects both A and $X - A$, then C intersects $\text{Bd } A$.

Problem 6

Let A be a proper subset of X , and let B be a proper subset of Y . If X and Y are connected, show that

$$(X \times Y) - (A \times B)$$

is connected. **Hint:** use the conclusion of Problem 3.

Problem 7

Let $\{X_\alpha\}_{\alpha \in J}$ be an indexed family of connected spaces; let X be the product space

$$X = \prod_{\alpha \in J} X_\alpha$$

Let $\mathbf{a} = (a_\alpha)$ be a fixed point of X .

- (a) Given any finite subset K of J , let X_K denote the subspace of X consisting of all points $\mathbf{x} = (x_\alpha)$ such that $x_\alpha = a_\alpha$ for $\alpha \notin K$. Show that X_K is connected.
- (b) Show that the union Y of the spaces X_K , for K finite subsets of J , is connected.
- (c) Show that X equals the closure of Y ; conclude that X is connected.

Problem 8

Let $Y \subset X$; let X and Y be connected. Show that if A and B form a separation of $X - Y$, then $Y \cup A$ and $Y \cup B$ are connected.

Problem 9

Let $f : S^1 \rightarrow \mathbb{R}$ be a continuous map. Show there exists a point $\mathbf{x}$ of S^1 such that $f(\mathbf{x}) = f(-\mathbf{x})$.

Problem 10

Show that if U is an open connected subspace of $\mathbb{R}^2$, then U is path connected. **Hint:** Show that given $x_0 \in U$, the set of points that can be joined to x_0 by a path in U is both open and closed in U .

Problem 11

If A is a connected subspace of X , does it follow that $\text{Int } A$ and $\text{Bd } A$ are connected? Does the converse hold? Justify your answers.